



## 9655 - Redshift identification of most distant radio quasar candidates at $z=7.4-9.5$

Cycle: 5, Proposal Category: GO

### INVESTIGATORS

| <i>Name</i>                             | <i>Institution</i>                                       |
|-----------------------------------------|----------------------------------------------------------|
| <b>Dr. Kohei Ichikawa (PI)</b>          | <b>Astronomical Institute, Tohoku University</b>         |
| Dr. Itsna Khoirul Fitriana (CoI) (CoPI) | National Astronomical Observatory of Japan (NAOJ)        |
| Dr. Xiaoyang Chen (CoI)                 | Tohoku University                                        |
| Dr. Kianhong Lee (CoI)                  | Nagoya University                                        |
| Dr. Takuya Hashimoto (CoI)              | University of Tsukuba                                    |
| Masayuki Akiyama (CoI)                  | Astronomical Institute, Tohoku University                |
| Dr. Yoshihiro Ueda (CoI)                | Kyoto University                                         |
| Dr. Tomoki Saito (CoI)                  | University of Hyogo                                      |
| Prof. Toshihiro Kawaguchi (CoI)         | University of Toyama                                     |
| Dr. Masatoshi Imanishi (CoI)            | National Astronomical Observatory of Japan (NAOJ)        |
| Dr. Bovornpratch Vijarnwannaluk (CoI)   | Academia Sinica, Institute of Astronomy and Astrophysics |
| Prof. Kotaro Kohno (CoI)                | University of Tokyo, Institute of Astronomy              |
| Dr. Kohei Inayoshi (CoI)                | Peking University                                        |
| Dr. Yoshiki Matsuoka (CoI)              | Ehime University                                         |
| Dr. Mariko Kubo (CoI)                   | Kwansei Gakuin University                                |
| Dr. Tohru Nagao (CoI)                   | Ehime University                                         |

### OBSERVATIONS

| <i>Folder</i> | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i>       | <i>Science Target</i> |
|---------------|--------------------|--------------|---------------------------------|-----------------------|
| VLASS-ID2     |                    |              |                                 |                       |
|               | 1                  | VLASS2_FS    | NIRSpec Fixed Slit Spectroscopy | (1) VLASS-ID2         |
|               | 2                  | VLASS12_FS   | NIRSpec Fixed Slit Spectroscopy | (2) VLASS-ID12        |
|               | 3                  | VLASS37_FS   | NIRSpec Fixed Slit Spectroscopy | (3) VLASS-ID37        |

| <b>Folder</b> | <b>Observation</b> | <b>Label</b> | <b>Observing Template</b>       | <b>Science Target</b> |
|---------------|--------------------|--------------|---------------------------------|-----------------------|
|               | 4                  | VLASS88_FS   | NIRSpec Fixed Slit Spectroscopy | (4) VLASS-ID88        |
|               | 5                  | VLASS94_FS   | NIRSpec Fixed Slit Spectroscopy | (5) VLASS-ID94        |

## ABSTRACT

The earliest ( $z > 8$ ) supermassive black holes hold unique information regarding the formation mechanism of the first generation black hole seeds. We propose to determine the redshifts of five radio quasar candidates at  $z_{\text{phot}} = 7.4\text{--}9.5$ , selected through the combination of deep Subaru/HSC optical, near-IR, and 3GHz VLASS radio surveys. These candidates exhibit significant 1.5GHz+3GHz radio emissions but lack emissions in the HSC optical (HSC-dark). Their emission starts to become evident in the J-band (referred to as y-dropouts). The IR color of these candidates suggests a dust-reddened feature with an estimated  $A_V$  of 1-5 mag, classifying them as prominent candidates of dust-reddened radio quasar at  $z = 7.4\text{--}9.5$ .

We propose to perform JWST/NIRSpec PRISM spectroscopy to cover the entire 1-5 $\mu\text{m}$  range, providing access to emission lines such as H $\beta$  (up to  $z < 9.5$ ) and MgII2800 ( $z < 16$ ), well known bright lines in quasars. Upon the confirmation of their redshifts, the candidates will become the most distant radio quasars at  $z = 7.4\text{--}9.5$ . Furthermore, their line width would enable us to estimate the BH masses of these targets. If these values align with expectations, they will give the significant tension to one of the current BH formation scenarios. If the values indicate under-massive compared to the ones assuming the Eddington-limit, it will make the first confirmation of super-Eddington accretion associated with radio emission in this early Universe at  $z = 8\text{--}9$ . The JWST/NIRSpec will serve as the unique and primary tool for determining spec- $z$  (and likely the BH mass estimates) for our targets, contributing the deeper understanding of SMBH formation and growth in the early Universe.

## OBSERVING DESCRIPTION

The proposed observations consist of NIRSpec PRISM spectroscopy for five "HSC-dark" VLASS radio quasar candidates. The targets are expected to be point sources considering their high- $z$  expectations of  $z = 7.4\text{--}9.5$ , and their J band AB magnitudes span between  $J = 24\text{--}25.5$ .

Our primary science goal is to obtain the spectroscopic redshift for these five targets, within the expected redshift range of  $z = 7.4\text{--}9.2$ . The wide spectral coverage of PRISM spectra spanning from 1 to 5  $\mu\text{m}$  contains multiple bright emission lines for the AGN within these expected redshifts, including H $\beta$ , [OIII]5007, MgII2800, and Ly $\alpha$ . Once the redshifts are confirmed, our second goal is to determine the line width of the expected broad emission lines, which are anticipated to reach widths greater than  $1.5 \times 10^3$  km/s. Even with the current PRISM's relatively low-resolution mode of  $R = 100\text{--}300$ , previous JWST/NIRSpec PRISM observations have demonstrated that such measurements of emission line width are applicable for AGN at redshifts ranging from 4 to 11.

We will allocate an integration time of 36.5 minutes for each source, resulting in a signal-to-noise ratio (SNR) exceeding 20 in the observed

wavelength range of 1  $\mu\text{m}$ . This SNR is sufficient to estimate the broad emission line characteristics as discussed above. To achieve this, we will utilize the S400A1 fixed slit.



# Proposal 9655 - Observation 1 - Redshift identification of most distant radio quasar candidates at z=7.4-9.5

|                   |                                                                                               |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
|-------------------|-----------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------|-----------------|------------|------------------|--------------------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation       | Proposal 9655, Observation 1: VLASS2_FS                                                       |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     | Mon Mar 16 22:00:14 GMT 2026 |                 |
|                   | Diagnostic Status: Warning                                                                    |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
|                   | Observing Template: NIRSpec Fixed Slit Spectroscopy                                           |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
| Diagnostics       | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.   |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
| Fixed Targets     | #                                                                                             | Name                     | Target Coordinates                                                                 |                 |            |                  | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                              |                 |
|                   | (1)                                                                                           | VLASS-ID2                | RA: 02 23 57.4428 (35.9893450d)<br>Dec: -05 26 40.33 (-5.44454d)<br>Equinox: J2000 |                 |            |                  | Epoch of Position: 2000  |                  |                    |                     |                              |                 |
|                   | Comments: J_vircam= 24.8434429mag_AB, H_vircam=23.0875435mag_AB, Ks_vircam=22.37317473mag_AB  |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
|                   | Category=Galaxy                                                                               |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
|                   | Description=[Active galactic nuclei, Quasars, Radio cores, Radio loud quasars]<br>Extended=NO |                          |                                                                                    |                 |            |                  |                          |                  |                    |                     |                              |                 |
| Acquisition       | #                                                                                             | Target                   | TA Method                                                                          | Subarray        | Filter     | Readout Pattern  | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                   | 1                                                                                             | SAME                     | WATA                                                                               | FULL            | F110W      | NRSRAPIDD6       | 3                        | 1                | 1                  | 171.788             | 173778.5                     |                 |
| Template          | HFF Readout Mode                                                                              |                          |                                                                                    |                 | Slit       |                  |                          |                  | Subarray           |                     |                              |                 |
|                   | false                                                                                         |                          |                                                                                    |                 | S400A1     |                  |                          |                  | FULL               |                     |                              |                 |
| Dithers           | #                                                                                             | Primary Dither Positions |                                                                                    |                 |            |                  |                          |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                   | 1                                                                                             | 3                        |                                                                                    |                 |            |                  |                          |                  | NONE               |                     |                              |                 |
| Spectral Elements | #                                                                                             | Grating/Filter           | Slit                                                                               | Readout Pattern | Groups/Int | Integrations/Exp | #                        | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                   | 1                                                                                             | PRISM/CLEAR              | S400A1                                                                             | NRSIRS2RAPID    | 50         | 1                | 1                        | NONE             | 3                  | 3                   | 2232.1                       | 173778.9        |

# Proposal 9655 - Observation 2 - Redshift identification of most distant radio quasar candidates at z=7.4-9.5

|                   |                                                                                             |                                                                                                                                                                                                    |                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|------------|------------------|--------------------------|------------------|--------------------|---------------------|---------------------|------------------------------|
| Observation       | Proposal 9655, Observation 2: VLASS12_FS                                                    |                                                                                                                                                                                                    |                                  |                 |            |                  |                          |                  |                    |                     |                     | Mon Mar 16 22:00:14 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                  |                                                                                                                                                                                                    |                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   | Observing Template: NIRSpec Fixed Slit Spectroscopy                                         |                                                                                                                                                                                                    |                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Diagnostics       | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                                                                                                                                    |                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Fixed Targets     | #                                                                                           | Name                                                                                                                                                                                               | Target Coordinates               |                 |            |                  | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                     |                              |
|                   | (2)                                                                                         | VLASS-ID12                                                                                                                                                                                         | RA: 16 11 49.8900 (242.9578750d) |                 |            |                  | Epoch of Position: 2000  |                  |                    |                     |                     |                              |
|                   |                                                                                             |                                                                                                                                                                                                    | Dec: +54 47 17.48 (54.78819d)    |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   |                                                                                             |                                                                                                                                                                                                    | Equinox: J2000                   |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   |                                                                                             | Comments: J_WFC=23.365147mag_AB, H_wfc=23.4233781mag_AB, K_wfc=22.568383mag_AB<br>Category=Galaxy<br>Description=[Active galactic nuclei, Quasars, Radio cores, Radio loud quasars]<br>Extended=NO |                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Acquisition       | #                                                                                           | Target                                                                                                                                                                                             | TA Method                        | Subarray        | Filter     | Readout Pattern  | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                   | 1                                                                                           | SAME                                                                                                                                                                                               | WATA                             | FULL            | F110W      | NRSRAPIDD6       | 3                        | 1                | 1                  | 171.788             | 173778.10           |                              |
| Template          | HFF Readout Mode                                                                            |                                                                                                                                                                                                    |                                  |                 | Slit       |                  |                          |                  | Subarray           |                     |                     |                              |
|                   | false                                                                                       |                                                                                                                                                                                                    |                                  |                 | S400A1     |                  |                          |                  | FULL               |                     |                     |                              |
| Dithers           | #                                                                                           | Primary Dither Positions                                                                                                                                                                           |                                  |                 |            |                  |                          |                  | Sub-Pixel Pattern  |                     |                     |                              |
|                   | 1                                                                                           | 3                                                                                                                                                                                                  |                                  |                 |            |                  |                          |                  | NONE               |                     |                     |                              |
| Spectral Elements | #                                                                                           | Grating/Filter                                                                                                                                                                                     | Slit                             | Readout Pattern | Groups/Int | Integrations/Exp | #                        | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | PRISM/CLEAR                                                                                                                                                                                        | S400A1                           | NRSIRS2RAPID    | 50         | 1                | 1                        | NONE             | 3                  | 3                   | 2232.1              | 173778.14                    |

# Proposal 9655 - Observation 3 - Redshift identification of most distant radio quasar candidates at z=7.4-9.5

|                   |                                                                                             |                                                                                                                                                                                |                                  |                 |            |                          |            |                   |                    |                     |                              |                 |
|-------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|------------|--------------------------|------------|-------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation       | Proposal 9655, Observation 3: VLASS37_FS                                                    |                                                                                                                                                                                |                                  |                 |            |                          |            |                   |                    |                     | Mon Mar 16 22:00:14 GMT 2026 |                 |
|                   | Diagnostic Status: Warning                                                                  |                                                                                                                                                                                |                                  |                 |            |                          |            |                   |                    |                     |                              |                 |
|                   | Observing Template: NIRSspec Fixed Slit Spectroscopy                                        |                                                                                                                                                                                |                                  |                 |            |                          |            |                   |                    |                     |                              |                 |
| Diagnostics       | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                                                                                                                |                                  |                 |            |                          |            |                   |                    |                     |                              |                 |
| Fixed Targets     | #                                                                                           | Name                                                                                                                                                                           | Target Coordinates               |                 |            | Targ. Coord. Corrections |            |                   | Miscellaneous      |                     |                              |                 |
|                   | (3)                                                                                         | VLASS-ID37                                                                                                                                                                     | RA: 16 16 55.6826 (244.2320108d) |                 |            | Epoch of Position: 2000  |            |                   |                    |                     |                              |                 |
|                   |                                                                                             |                                                                                                                                                                                | Dec: +55 01 1.51 (55.01709d)     |                 |            |                          |            |                   |                    |                     |                              |                 |
|                   |                                                                                             |                                                                                                                                                                                | Equinox: J2000                   |                 |            |                          |            |                   |                    |                     |                              |                 |
|                   |                                                                                             | Comments: J_WFC=23.5893822 mag_AB, K_wfc=22.0246391 mag_AB<br>Category=Galaxy<br>Description=[Active galactic nuclei, Quasars, Radio cores, Radio loud quasars]<br>Extended=NO |                                  |                 |            |                          |            |                   |                    |                     |                              |                 |
| Acquisition       | #                                                                                           | Target                                                                                                                                                                         | TA Method                        | Subarray        | Filter     | Readout Pattern          | Groups/Int | Integrations/Exp  | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                   | 1                                                                                           | SAME                                                                                                                                                                           | WATA                             | FULL            | F110W      | NRSRAPIDD6               | 3          | 1                 | 1                  | 171.788             | 173778.11                    |                 |
| Template          | HFF Readout Mode                                                                            |                                                                                                                                                                                |                                  | Slit            |            |                          |            | Subarray          |                    |                     |                              |                 |
|                   | false                                                                                       |                                                                                                                                                                                |                                  | S400A1          |            |                          |            | FULL              |                    |                     |                              |                 |
| Dithers           | #                                                                                           | Primary Dither Positions                                                                                                                                                       |                                  |                 |            |                          |            | Sub-Pixel Pattern |                    |                     |                              |                 |
|                   | 1                                                                                           | 3                                                                                                                                                                              |                                  |                 |            |                          |            | NONE              |                    |                     |                              |                 |
| Spectral Elements | #                                                                                           | Grating/Filter                                                                                                                                                                 | Slit                             | Readout Pattern | Groups/Int | Integrations/Exp         | #          | Autocal           | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                   | 1                                                                                           | PRISM/CLEAR                                                                                                                                                                    | S400A1                           | NRSIRS2RAPID    | 50         | 1                        | 1          | NONE              | 3                  | 3                   | 2232.1                       | 173778.15       |

# Proposal 9655 - Observation 4 - Redshift identification of most distant radio quasar candidates at z=7.4-9.5

|                   |                                                                                               |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
|-------------------|-----------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------|-----------------|------------|--------------------------|------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation       | Proposal 9655, Observation 4: VLASS88_FS                                                      |                          |                                                                                    |                 |            |                          |            |                  |                    |                     | Mon Mar 16 22:00:14 GMT 2026 |                 |
|                   | Diagnostic Status: Warning                                                                    |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
|                   | Observing Template: NIRSspec Fixed Slit Spectroscopy                                          |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
| Diagnostics       | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.   |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
| Fixed Targets     | #                                                                                             | Name                     | Target Coordinates                                                                 |                 |            | Targ. Coord. Corrections |            |                  | Miscellaneous      |                     |                              |                 |
|                   | (4)                                                                                           | VLASS-ID88               | RA: 10 00 16.5701 (150.0690421d)<br>Dec: +02 26 38.41 (2.44400d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |            |                  |                    |                     |                              |                 |
|                   | Comments: J_vircam= 25.54mag_AB, H_vircam=23mag_AB, Ks_vircam=21.94mag_AB                     |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
|                   | Category=Galaxy                                                                               |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
|                   | Description=[Active galactic nuclei, Quasars, Radio cores, Radio loud quasars]<br>Extended=NO |                          |                                                                                    |                 |            |                          |            |                  |                    |                     |                              |                 |
| Acquisition       | #                                                                                             | Target                   | TA Method                                                                          | Subarray        | Filter     | Readout Pattern          | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                   | 1                                                                                             | SAME                     | WATA                                                                               | FULL            | F110W      | NRSRAPIDD6               | 3          | 1                | 1                  | 171.788             | 173778.12                    |                 |
| Template          | HFF Readout Mode                                                                              |                          |                                                                                    |                 | Slit       |                          |            |                  | Subarray           |                     |                              |                 |
|                   | false                                                                                         |                          |                                                                                    |                 | S400A1     |                          |            |                  | FULL               |                     |                              |                 |
| Dithers           | #                                                                                             | Primary Dither Positions |                                                                                    |                 |            |                          |            |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                   | 1                                                                                             | 3                        |                                                                                    |                 |            |                          |            |                  | NONE               |                     |                              |                 |
| Spectral Elements | #                                                                                             | Grating/Filter           | Slit                                                                               | Readout Pattern | Groups/Int | Integrations/Exp         | #          | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                   | 1                                                                                             | PRISM/CLEAR              | S400A1                                                                             | NRSIRS2RAPID    | 50         | 1                        | 1          | NONE             | 3                  | 3                   | 2232.1                       | 173778.16       |

Proposal 9655 - Observation 4 - Redshift identification of most distant radio quasar candidates at  $z=7.4-9.5$

|                      |                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Aperture PA Range 40 to 110 Degrees (V3 261.135849 to 331.135849)<br>Aperture PA Range 220 to 290 Degrees (V3 81.135849 to 151.135849) |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 9655 - Observation 5 - Redshift identification of most distant radio quasar candidates at z=7.4-9.5

|                   |                                                                                             |                          |                                                                                                                                                                                                 |                 |            |                  |                          |                  |                    |                     |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|------------------|--------------------------|------------------|--------------------|---------------------|---------------------|------------------------------|
| Observation       | Proposal 9655, Observation 5: VLASS94_FS                                                    |                          |                                                                                                                                                                                                 |                 |            |                  |                          |                  |                    |                     |                     | Mon Mar 16 22:00:14 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                  |                          |                                                                                                                                                                                                 |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   | Observing Template: NIRSpec Fixed Slit Spectroscopy                                         |                          |                                                                                                                                                                                                 |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Diagnostics       | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                          |                                                                                                                                                                                                 |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Fixed Targets     | #                                                                                           | Name                     | Target Coordinates                                                                                                                                                                              |                 |            |                  | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                     |                              |
|                   | (5)                                                                                         | VLASS-ID94               | RA: 10 00 6.1704 (150.0257100d)                                                                                                                                                                 |                 |            |                  | Epoch of Position: 2000  |                  |                    |                     |                     |                              |
|                   |                                                                                             |                          | Dec: +02 40 0.47 (2.66680d)                                                                                                                                                                     |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   |                                                                                             |                          | Equinox: J2000                                                                                                                                                                                  |                 |            |                  |                          |                  |                    |                     |                     |                              |
|                   |                                                                                             |                          | Comments: J_vircam= 24.73mag_AB, H_vircam=23.11mag_AB, Ks_vircam=22.3mag_AB<br>Category=Galaxy<br>Description=[Active galactic nuclei, Quasars, Radio cores, Radio loud quasars]<br>Extended=NO |                 |            |                  |                          |                  |                    |                     |                     |                              |
| Acquisition       | #                                                                                           | Target                   | TA Method                                                                                                                                                                                       | Subarray        | Filter     | Readout Pattern  | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID     |                              |
|                   | 1                                                                                           | SAME                     | WATA                                                                                                                                                                                            | FULL            | F110W      | NRSRAPIDD6       | 3                        | 1                | 1                  | 171.788             | 173778.13           |                              |
| Template          | HFF Readout Mode                                                                            |                          |                                                                                                                                                                                                 |                 | Slit       |                  |                          |                  | Subarray           |                     |                     |                              |
|                   | false                                                                                       |                          |                                                                                                                                                                                                 |                 | S400A1     |                  |                          |                  | FULL               |                     |                     |                              |
| Dithers           | #                                                                                           | Primary Dither Positions |                                                                                                                                                                                                 |                 |            |                  |                          |                  | Sub-Pixel Pattern  |                     |                     |                              |
|                   | 1                                                                                           | 3                        |                                                                                                                                                                                                 |                 |            |                  |                          |                  | NONE               |                     |                     |                              |
| Spectral Elements | #                                                                                           | Grating/Filter           | Slit                                                                                                                                                                                            | Readout Pattern | Groups/Int | Integrations/Exp | #                        | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | PRISM/CLEAR              | S400A1                                                                                                                                                                                          | NRSIRS2RAPID    | 50         | 1                | 1                        | NONE             | 3                  | 3                   | 2232.1              | 173778.16                    |